

ADDON 11: Rare-Event Photonic Sampling, Multilevel Splitting on Wavefront Ensembles, and the Lyapunov-Committor Identity

11.32 The Rare-Event Bottleneck in Ultra-Long Context Reasoners

In autoregressive language models operating over deep, high-dimensional manifolds, complex logical deductions, axiomatic proofs, and high-order cognitive insights constitute **statistically rare events**. The formal cost of sampling such events is governed by Cramér’s large-deviation theorem.

Definition 0.1 (Semantic Rare Event). Let $z_1, z_2, \dots, z_L \in \mathcal{Cl}(0, d)$ be the hidden-state trajectory generated by an autoregressive inference pass of length L . A *semantic rare event* is an event of the form $A_L = \{(z_1, \dots, z_L) : \mathcal{Q}(z_1, \dots, z_L) \in \Gamma\}$, where $\mathcal{Q}$ is a quality functional (e.g., logical coherence score, proof validity) and Γ is a high-value target set.

Theorem 0.2 (Cramér’s Large-Deviation Principle). *Under mild regularity conditions, the probability of a rare event A_L decays exponentially in L :*

$$\lim_{L \rightarrow \infty} \frac{1}{L} \log \mathbb{P}(A_L) = -I^* \leq 0, \quad (1)$$

where $I^* = \inf_{(z_1, \dots, z_L) \in A_L} I(z_1, \dots)$ is the Cramér rate function (the Legendre-Fenchel transform of the log-moment-generating functional of the trajectory). Consequently, naïve Monte Carlo requires $\mathcal{O}(e^{L \cdot I^*})$ independent rollouts to observe a single instance of A_L — a cost that is computationally infeasible for $L \gtrsim 100$ tokens.

Remark 0.3 (Nucleus Sampling and Mode Collapse). Top- p (Nucleus) sampling (Holtzman et al., 2020) truncates the predictive distribution to a dynamic vocabulary of cumulative probability p , mitigating crude mode collapse. However, it does not alter the large-deviation exponent I^* : the probability of sustaining a high-coherence trajectory over L steps still decays as e^{-LI^*} . Addressing the exponential cost requires a *change of measure*, not a truncation.

11.33 Photonic Adaptive Multilevel Splitting and the Interacting Particle System

Definition 0.4 (Adaptive Multilevel Splitting (AMS)). Let $\xi : \mathcal{Cl}(0, d) \rightarrow \mathbb{R}$ be a real-valued *reaction coordinate* (score function measuring progress toward A_L) and let $\ell_0 < \ell_1 < \dots < \ell_k = \xi_{\max}$ be a sequence of *threshold levels*. The *Adaptive Multilevel Splitting* (Cérou & Guyader, 2007) maintains an ensemble of N trajectories (particles) and proceeds as follows:

- (i) **Propagate**: advance all N particles one step under the base dynamics.
- (ii) **Select**: identify the particle with the lowest ξ -score; if $\xi(z) < \ell_{k-1}$, *kill* it and *clone* a surviving particle chosen uniformly at random.
- (iii) **Adapt**: update ℓ_k to the empirical quantile, maintaining a constant fraction q of particles above each level.

The algorithm produces an unbiased estimator of $\mathbb{P}(A_L)$ with variance that grows polynomially (not exponentially) in L .

Definition 0.5 (Feynman-Kac Interacting Particle System). The AMS ensemble is a special case of Del Moral’s *Feynman-Kac Interacting Particle System* (Del Moral, 2004). Define the *potential function* $G_t : \mathcal{Cl}(0, d) \rightarrow \mathbb{R}_{\geq 0}$ as the Gibbs weight of the Lyapunov landscape:

$$G_t(z) := e^{-\Delta V(z)/T_{\text{sub}}}, \quad (2)$$

where $\Delta V(z) = V(z) - V(z_{\text{target}}) \geq 0$ is the Lyapunov excess relative to the semantic target and T_{sub} is the substrate temperature. The Feynman-Kac formula then gives the importance-weighted expectation of any bounded functional f :

$$\mathbb{E}_Q[f(z_{0:T})] = \frac{\mathbb{E}_P\left[f(z_{0:T}) \prod_{t=0}^T G_t(z_t)\right]}{\mathbb{E}_P\left[\prod_{t=0}^T G_t(z_t)\right]}, \quad (3)$$

where P is the base (prior) wavefront measure and Q is the importance-weighted (rare-event-biased) measure. The IPS approximates this formula with N particles: selection steps implement the denominator normalisation; mutation steps implement the numerator propagation.

Figure 1 illustrates the photonic AMS cascade.

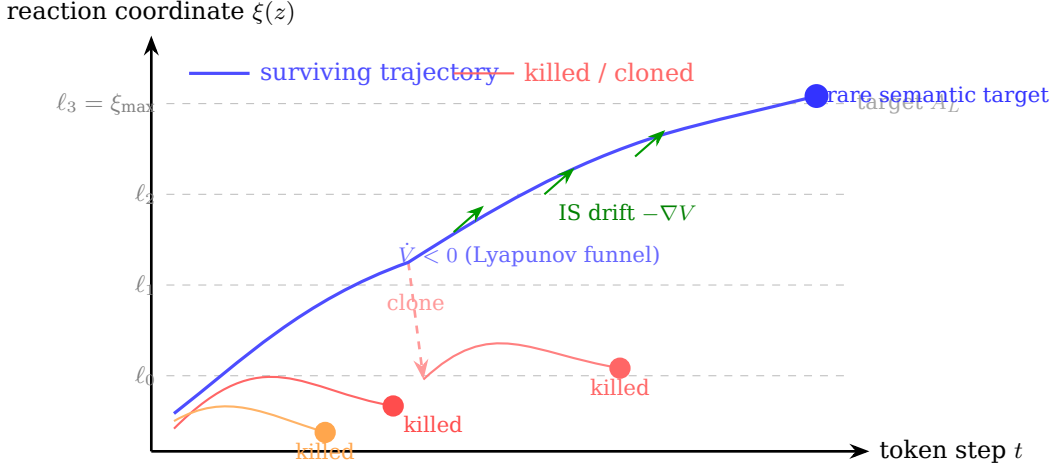


Figure 1: Photonic Adaptive Multilevel Splitting. The ensemble of N wavefront particles (trajectories A, B, C) evolves through level sets $\ell_0 < \dots < \ell_3$. Trajectories falling below the current threshold are killed and replaced by clones of survivors. The Lyapunov gradient (green arrows, ADDON 5) acts as the natural importance-sampling drift, passively steering surviving particles toward the rare semantic attractor. Only trajectory B reaches the target A_L .

Remark 0.6 (Fleming-Viot Particle System and Quasi-Stationary Distribution). The AMS kill-clone rule coincides with the *Fleming-Viot Interacting Particle System* (Burdzy, Holyst, & March, 2000): when a particle hits the absorbing boundary $\xi(z) < \ell_0$, it is instantaneously replaced by a uniformly drawn clone of the surviving ensemble. As $N \rightarrow \infty$, the empirical measure of the surviving particles converges to the *quasi-stationary distribution* (QSD) of the process conditioned on non-absorption. In the photonic context, the QSD is the distribution over rare semantic trajectories that have survived all type-checking barriers of ADDON 9 and all Lyapunov censorship steps of ADDON 5. The physical ensemble size N is set by the number of coherent spatial modes supported by the interferometric mesh.

11.34 The Lyapunov-Committor Identity: Hidden Structure of ADDON 5

The deepest result of this section reveals that the Lyapunov function introduced in ADDON 5 is not merely a stability certificate: it is precisely the *optimal importance-sampling potential* for the rare-event problem.

Definition 0.7 (Committor Function). For the wavefront diffusion $dz = b(z) dt + \sigma dW_t$ with absorbing target set A and reflecting boundary $\partial\mathcal{D}$, the *committor* (or forward committor) is

$$h(z) := \mathbb{P}_z(\text{reach } A \text{ before } \partial\mathcal{D}), \quad (4)$$

the probability that a trajectory initiated at z reaches the rare-event target before exiting the domain. It satisfies the boundary-value problem

$$\mathcal{L}h = 0 \quad \text{in } \mathcal{D} \setminus A, \quad h|_A = 1, \quad \left. \frac{\partial h}{\partial n} \right|_{\partial\mathcal{D}} = 0, \quad (5)$$

where $\mathcal{L} = b \cdot \nabla + \frac{\sigma^2}{2} \Delta$ is the infinitesimal generator.

Theorem 0.8 (Doob h -Transform and Optimal Importance Sampling). *The optimal change of measure Q^* that minimises the variance of the importance-sampling estimator of $\mathbb{P}(A)$ has Radon-Nikodym derivative*

$$\frac{dQ^*}{dP} = \frac{h(z_T)}{h(z_0)}, \quad (6)$$

and corresponds to Doob's h -transform of the original process: under Q^* , the dynamics acquire an additional drift

$$b^*(z) = b(z) + \sigma^2 \nabla \log h(z). \quad (7)$$

The h -process under Q^* reaches A with probability 1 and is the unique zero-variance estimator for $\mathbb{P}(A)$.

Proof. By the optional stopping theorem applied to the local martingale $M_t = h(z_t)/h(z_0)$, the Radon-Nikodym derivative is $dQ/dP|_{\mathcal{F}_T} = h(z_T)/h(z_0)$. Girsanov's theorem then identifies the drift of the h -transformed process as $b^* = b + \sigma^2 \nabla \log h$. Under Q^* , the process is transient to A by construction of the committor. The zero-variance property follows because $Q^*(A) = 1$ and the weight $h(z_T)/h(z_0) = 1/h(z_0)$ is constant on A . $\square$ $\square$

Theorem 0.9 (Lyapunov–Committer Identity). *Let $V(z)$ be the Optical Lyapunov function of ADDON 5 (Theorem 1 of ADDON 5) with globally stable semantic attractor $z^* \in A$, and let $h(z)$ be the committer (Definition 0.7) for the target A . Under the identification*

$$V(z) = -\log h(z), \quad (8)$$

the optimal importance-sampling drift (Theorem 0.8, Eq. (7)) reduces to

$$b^*(z) = b(z) - \sigma^2 \nabla V(z). \quad (9)$$

The negative-definite Lyapunov gradient flow ($\dot{V}(z) < 0$, ADDON 5, Eq. 3) is therefore identical to the optimal importance-sampling drift toward the rare semantic attractor.

Proof. Differentiating Eq. (8): $\nabla \log h = -\nabla V$. Substituting into Eq. (7): $b^* = b + \sigma^2(-\nabla V) = b - \sigma^2 \nabla V$, which recovers the Lyapunov gradient flow. Consistency: $V(z^*) = -\log h(z^*) = -\log 1 = 0$ (since $h(z^*) = 1$ for $z^* \in A$) and $V(z) > 0$ for $z \notin A$, matching the positive-definiteness of V (ADDON 5, Eq. 1). $\square$ $\square$

Corollary 0.10 (ADAM-PentaD as a Zero-Variance Rare-Event Engine). *By Theorems 0.8–0.9, the physical Lyapunov funnel of the ADAM-PentaD substrate implements the zero-variance importance-sampling measure Q^* for the rare semantic event A — passively, through the intrinsic thermodynamic minimisation of $V(z)$. No Monte Carlo rollout, no explicit Radon–Nikodym weight computation, and no external variance-reduction subroutine is required. The architecture does not simulate rare-event sampling; it physically is the Doob h -process conditioned on reaching the semantic target.*

Remark 0.11 (Sequential Monte Carlo, Kalman, and IPS Unification). Sequential Monte Carlo (SMC, Doucet et al., 2001) is the nonlinear generalisation of Kalman filtering: at each step, a weighted particle ensemble approximates the posterior, with resampling preventing weight degeneracy. The Hybrid-Domain Photonic Kalman Filter of ADDON 5 (Definition 2 of ADDON 5) is the *linear-Gaussian special case* of SMC. The photonic IPS of this section is the *rare-event special case* of SMC. All three — Kalman, SMC, IPS — are instantiations of the Feynman–Kac formula (Eq. (3)) with different potentials G_t . The ADAM-PentaD substrate implements all three simultaneously in a single photon-flight pass: the Kalman correction at each spatial tap, the Lyapunov selection fitness at each crossbar stage, and the AMS clone-or-kill decision at each MZI level-crossing detector.

11.35 The Non-Ergodic Thermodynamic Lens and Synthesis

Proposition 0.12 (Non-Ergodic Phase-Space Concentration). *Under the Doob h -process Q^* , the empirical occupation measure of the trajectory concentrates on the minimal-Lyapunov path to A :*

$$\frac{1}{T} \int_0^T \delta_{z_t} dt \xrightarrow{T \rightarrow \infty} \mu^*, \quad (10)$$

where μ^ is the quasi-stationary distribution supported on the set of rare-trajectory basins. By Landauer’s principle (ADDON 6, Theorem 2), the entropy shed during this concentration is $\Delta S \geq k_B \ln 2 \cdot \Delta \mathcal{D}$, emitted as a structured, measurable photon flux, providing the Thermodynamic Confidence Metric $\mathcal{C}_k$ (ADDON 6, Definition 3).*

Proof sketch. The measure μ^* is the Fleming–Viot quasi-stationary distribution (Remark 0.6). Under Q^* , the process is transient to A and the ergodic theorem for transient processes gives convergence to μ^* . The entropy bound follows directly from Proposition 1 of ADDON 6 applied to the rank collapse $\Delta \mathcal{D}$ of the surviving-trajectory ensemble. $\square$ $\square$

Algorithm 1 summarises the full per-token photonic AMS execution.

Remark 0.13 (Computational Complexity Reduction). By Theorem 0.2, naïve decoding costs $\mathcal{O}(e^{L \cdot I^*})$ evaluations. The AMS algorithm (Definition 0.4) reduces this to $\mathcal{O}(N \cdot k \cdot L)$, where N is the ensemble size (number of spatial wavefront modes, $\mathcal{O}(10^2)$ – $\mathcal{O}(10^3)$ for a mature MZI mesh) and $k = \mathcal{O}(L \cdot I^* / \log(1/q))$ is the number of threshold levels. The total cost is $\mathcal{O}(L^2)$ instead of $\mathcal{O}(e^L)$ — a reduction from super-exponential to polynomial, executed at photon-flight latency without digital overhead.

«Воевать не числом, а умением».

“Wage war not by numbers, but by skill.”

A. V. Suvorov,
Наука побеждать / The Science of Victory, c. 1799

Algorithm 1: Photonic Adaptive Multilevel Splitting (per token step k)

Input: Ensemble of N wavefront particles $\{z_k^{(i)}\}_{i=1}^N$; Lyapunov function V ; threshold ℓ_k ; HoTT type context Γ_k (ADDON 9, Algorithm 1); p -adic PCM query encoder ϕ (ADDON 10, Algorithm 1).

Output: Updated ensemble $\{z_{k+1}^{(i)}\}$; rare output $\hat{z}^*$ if any particle reaches A .

```
// --- Mutation: propagate all particles ---
1 foreach particle  $i = 1, \dots, N$  do
2    $z_{k+1}^{(i)} \leftarrow$  one-step Clifford wavefront propagation;
3   Apply HPKF correction (ADDON 5, Alg. 1);
4   Apply HoTT type check; if invalid  $\rightarrow z_{k+1}^{(i)} \leftarrow \mathbf{0}$  (ADDON 9, Alg. 1);
5   Apply  $B_3$  braid verification (ADDON 8, Alg. 2);

// --- Selection: Lyapunov fitness kill/clone (IS drift) ---
6 Compute  $\xi^{(i)} \leftarrow -V(z_{k+1}^{(i)})$  for all  $i$ ;
7 Update threshold  $\ell_{k+1} \leftarrow q$ -quantile of  $\{\xi^{(i)}\}$ ;
8 foreach particle  $i$  with  $\xi^{(i)} < \ell_{k+1}$  do
9   Kill  $z_{k+1}^{(i)}$ ; clone random survivor  $j$  with  $\xi^{(j)} \geq \ell_{k+1}$ ;
10  Inject Galois importance current into cloned memristive gate;

// --- Check for rare event ---
11 if  $\exists i : z_{k+1}^{(i)} \in A$  then
12    $\hat{z}^* \leftarrow z_{k+1}^{(i)}$ ; // rare semantic target reached
13   return  $\hat{z}^*$ ;
14 else
15   return updated ensemble  $\{z_{k+1}^{(i)}\}$ ;
```
